

* Chi Square test:

-> Chi Square test for goodness of fit test claims about population proportions.

-> It is a non parametric test that is performed on categorical (ordinal and nominal) data.

Ex: There is a research that claimed that yellow, red & orange bikes were liked in equal probabilities. Now that study is tested by collecting sample

	Theory	Sample	
yellow	$\frac{1}{3}$	22	→ Perform goodness of fit of sample to theory.
Red	$\frac{1}{3}$	17	
Orange	$\frac{1}{3}$	59	
	↓	↓	
	Categorical dist	Observed Categorical dist	

Ex:

In a science class of 75 students, 11 are left handed. Does this class fit the theory that 12% of students are left handed.

	Observed	Expected	
Left	11	9	$\frac{3}{100} \times 75 = 9$
Right	64	66	

1. In 2010 census of the city, the weight of the individuals in a small city were found to be the following

< 50kg	50-75	> 75
20%	30%	50%

In 2020, weight of n=500 individuals were sampled. Below are the results

< 50kg	50-75	> 75
140	160	200

Using $\alpha=0.05$, would you conclude the population differences of weights

Sol.

<50kg	50-75	>75
20%	30%	50%

2010
Expected data

<50kg	50-75	>75
140	160	200

2020 $n = 500$
Observed data

<50kg	50-75	>75
0.2×500 $= 100$	0.3×500 $= 150$	0.5×500 $= 250$

Expected data

① H_0 : The data meets the expectation

② H_1 : The data doesn't meet the expectation

• $\alpha = 0.05$

$C\alpha = 1 - 0.05$

$= 95\%$

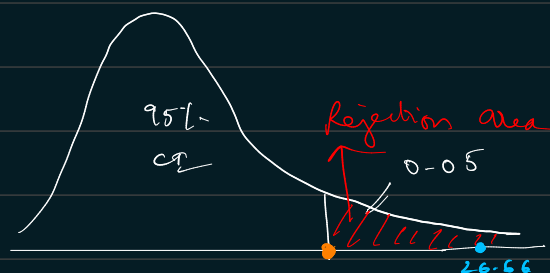
③ Degree of freedom, $df = k - 1$

$k \rightarrow$ no. of Categories

$= 3 - 1 = 2$

④ Decision boundary:-

$\rightarrow$ Z & t test has symmetrical dist., not this



Critical value = 5.991 (from χ^2 table across
dof & α)

Critical value = 5.991

If $\chi^2 > 5.991 \rightarrow$ Reject H_0

5) χ^2 test statistic :-

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

$$= \frac{(140 - 100)^2}{100} + \frac{(160 - 150)^2}{150} + \frac{(200 - 250)^2}{250}$$

$$= 16 + 0.66 + 10$$

$$= 26.66 > 5.991$$

$\Rightarrow$ Reject H_0

The weights of 2020 pop. is different from expected in the 2010 pop.

<50kg	50-75	>75
140	160	200

O

<50kg	50-75	>75
0.2×500	0.3×500	0.5×500
$= 100$	$= 150$	$= 250$

E

* Analysis of Variance (ANOVA) :-

-> Anova is a statistical method used to compare the means of two or more groups

* Components of ANOVA:-

1. Factors (Variables)
2. Levels

* Assumptions of ANOVA:-

① Normality of sampling dist. of mean

The dist. of sample mean is normally distributed [Central theorem]

② Absence of outliers :-

Outlying score should be removed from dataset

③ Homogeneity of variance

Population variance in different levels of each independent variable should be equal

$$[\sigma_1^2 = \sigma_2^2 = \sigma_3^2]$$

④ Samples are independent and random

* Types of ANOVA:-

① One way ANOVA:-

One factor with atleast 2 levels, these levels are independent

Ex:- Doctor wants to test a new medication & dosage-

Medication → Factor			↔ Levels
10 mg	20 mg	30 mg	
5	6	3	
2	7	5	
—	—	—	
—	—	—	
—	—	—	

② Repeated measures ANOVA:-

One factor with atleast 2 levels. Levels are dependent.

Ex: Individual running performance (daily)

Running → Factor		
Day 1	Day 2	Day 3 → levels are dependent
2	6	7
3	5	8
—	—	—
←	←	←

③ Factorial ANOVA:-

Two or more factors each of which with atleast 2 levels.

Levels can be independent or dependent.

Ex:

		Running → Factor 1		
		Day 1	Day 2	Day 3 → levels (dependent)
Gender →	male	2	6	8
		3	5	7
Factor 2 ↓	female	1	6	3
		7	8	5
		↓ levels (independent)		

* Hypothesis testing in ANOVA (Partitioning of Variance in ANOVA) :-

① Null hypothesis :-

$$H_0 : \mu_1 = \mu_2 = \mu_3 \dots$$

② Alternate hypothesis :-

H_1 : At least 1 of the sample mean is not equal

③ Test statistic (F test) :-

$$F = \frac{\text{Variance b/w samples}}{\text{Variance within samples}}$$

1. One way ANOVA

Doctors want to test a new medication which reduces headache. They split the participants into 3 groups [15mg, 30mg, 45mg]. Later the doctor asks the patients to rate the headache b/w [1- 10]. Are there any differences between the 3 groups using $\alpha = 0.05$?

15 mg	30 mg	45 mg
9	7	4
8	6	3
7	6	2
8	7	3
8	8	4
9	7	3
8	6	2

Sol ① $H_0 : \mu_{15} = \mu_{30} = \mu_{45}$

② H_1 : not all means are equal

③ Significance, $\alpha = 0.05$
 $cr = 95\%$

④ Degr of freedom :-

$$n = 7$$

$$a = 3$$

$$N = 7 \times 3 = 21 \quad (\text{Total sample})$$

$$\text{df}_{\text{b/w}} = a - 1$$

$$\rightarrow df_1$$

$$\text{df}_{\text{b/w}} = 3 - 1 = 2$$

$$\text{df}_{\text{within}} = N - a$$

$$\rightarrow df_2$$

$$\text{df}_{\text{within}} = 21 - 3 = 18$$

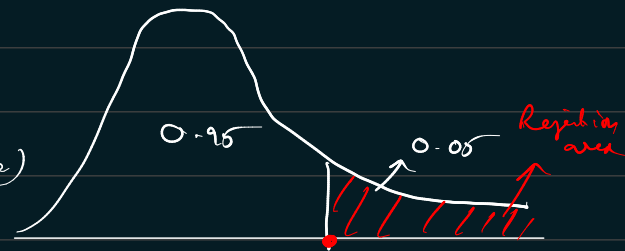
$$\text{df}_{\text{total}} = N - 1$$

$$\text{df}_{\text{total}} = 21 - 1 = 20$$

⑤ Decision boundary :-

$$\text{critical value} = 3.55$$

(against df_1, df_2 & α in F table)



$$\text{If } F_{\text{cal}} \geq 3.55 \Rightarrow \text{Reject } H_0$$

$$CV = 3.55$$

⑥ F test statistic :-

$$F = \frac{\text{Variance b/w samples}}{\text{Variance within samples}}$$

	SS	df	MS	F
b/w	98.67	2	59.34	

within	10.29	18	0.57	86.56
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Total	108.96	20	59.88	
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15 mg	30 mg	45 mg
9	7	4
8	6	3
7	6	2
8	7	3
8	8	4
9	7	3
8	6	2

a) Sum of Squares b/w :-

$$\frac{\sum (\sum a_i)^2}{n} - \frac{T^2}{N}$$

15mg :- $4 + 8 + 7 + 8 + 8 + 9 + 8 = 52$

30mg :- $7 + 6 + 6 + 7 + 8 + 7 + 6 = 47$

45mg :- $4 + 3 + 2 + 3 + 4 + 3 + 2 = 21$

$$\Rightarrow \frac{52^2 + 47^2 + 21^2}{7} - \left[\frac{52^2 + 47^2 + 21^2}{21} \right]$$

$$= 98.67$$

b) Sum of Squares within

$$\sum Y^2 - \frac{\sum (\sum a_i)^2}{n}$$

$$\sum Y^2 = 9^2 + 8^2 + 7^2 + 8^2 + \dots$$

$$= 853$$

15 mg	30 mg	45 mg
9	7	4
8	6	3
7	6	2
8	7	3
8	8	4
9	7	3
8	6	2

$$\Rightarrow 853 - \frac{52^2 + 47^2 + 21^2}{7}$$

$$= 10.21$$

c) Mean Square b/w

$$\frac{SS}{df}$$

$$\frac{98.67}{2} = 49.34$$

d) Mean Square within

$$\frac{SS}{df}$$

$$\frac{10.29}{18} = 0.54$$

c)

$$F_{\text{test}} = \frac{MS_{\text{b/w}}}{MS_{\text{within}}} = F = \frac{\text{Variance b/w samples}}{\text{Variance within samples}}$$

$$F_{\text{test}} = \frac{49.34}{0.54} = 86.56 > 3.55$$

If $F_{\text{cal}} \geq 3.55 \Rightarrow \text{Reject } H_0$

$\Rightarrow$ Reject H_0